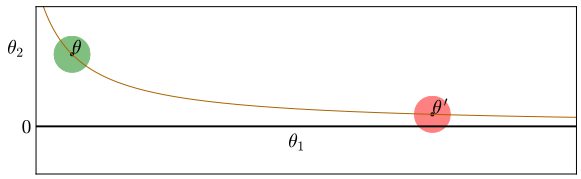


Figure 4: An illustration of how we exploit non-identifiability and its particular geometry to obtain sharper minima: although θ is far from the $\theta_2 = 0$ line, the observationally equivalent parameter θ' is closer. The green and red circle centered on each of these points have the same radius. Best seen with colors.

define $\alpha = \frac{\epsilon}{\|\theta_1\|_2}$. We will now consider the observationally equivalent parameter $T_\alpha(\theta_1, \theta_2) = (\epsilon \frac{\theta_1}{\|\theta_1\|_2}, \alpha^{-1}\theta_2)$. Given that $\|\theta_1\|_2 \leq \|\theta\|_2$, we have that $(0, \alpha^{-1}\theta_2) \in B_2\big(\epsilon, T_\alpha(\theta)\big)$, making the maximum loss in this neighborhood at least as high as the best constant-valued function, incurring relatively high sharpness. Figure 4 provides a visualization of the proof.

For rectified neural network every minimum is observationally equivalent to a minimum that generalizes as well but with high ϵ-sharpness. This also applies when using the *full-space* ϵ-sharpness used by Keskar et al. (2017). We can prove this similarly using the equivalence of norms in finite dimensional vector spaces and the fact that for $c > 0, \epsilon > 0, \epsilon \leq \epsilon(c+1)$ (see Keskar et al. (2017)). We have not been able to show a similar problem with *random subspace* ϵ-sharpness used by Keskar et al. (2017), i.e. a restriction of the maximization to a random subspace, which could relate to the notion of *wide valleys* described by Chaudhari et al. (2017).

By exploiting the non-Euclidean geometry and non-identifiability of rectified neural networks, we were able to demonstrate some of the limits of using typical definitions of minimum's flatness as core explanation for generalization.

5 Allowing reparametrizations

In the previous section 4 we explored the case of a fixed parametrization, that of deep rectifier models. In this section we demonstrate a simple observation. If we are allowed to change the parametrization of some function f, we can obtain arbitrarily different geometries without affecting how the function evaluates on unseen data. The same holds for reparametrization of the input space. The implication is that the correlation between the geometry of the parameter space (and hence the error surface) and the behavior of a given function is meaningless if not preconditioned on the specific parametrization of the model.

5.1 Model reparametrization

One thing that needs to be considered when relating flatness of minima to their probable generalization is that the choice of parametrization and its associated geometry are arbitrary. Since we are interested in finding a prediction function in a given family of functions, no reparametrization of this family should influence generalization of any of these functions. Given a bijection g onto θ, we can define new transformed parameter $\eta = g^{-1}(\theta)$. Since θ and η represent in different space the same prediction function, they should generalize as well.

Let's call $L_\eta = L \circ g$ the loss function with respect to the new parameter η. We generalize the derivation of Subsection 4.2:

$$\begin{aligned} L_\eta(\eta) &= L\big(g(\eta)\big) \\ \Rightarrow (\nabla L_\eta)(\eta) &= (\nabla L)\big(g(\eta)\big)(\nabla g)(\eta) \\ \Rightarrow (\nabla^2 L_\eta)(\eta) &= (\nabla g)(\eta)^T (\nabla^2 L)\big(g(\eta)\big)(\nabla g)(\eta) \\ &\quad + (\nabla L)\big(g(\eta)\big)(\nabla^2 g)(\eta). \end{aligned}$$

At a differentiable critical point, we have by definition $(\nabla L)\big(g(\eta)\big) = 0$, therefore the transformed Hessian at a critical point becomes

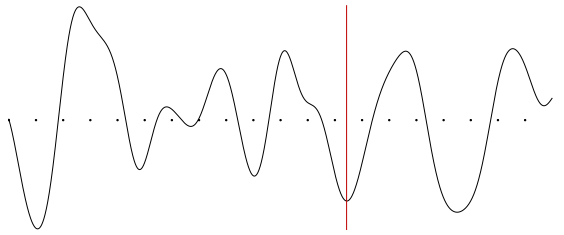
$$(\nabla^2 L_\eta)(\eta) = (\nabla g)(\eta)^T (\nabla^2 L)\big(g(\eta)\big)(\nabla g)(\eta).$$

This means that by reparametrizing the problem we can modify to a large extent the geometry of the loss function so as to have sharp minima of L in θ correspond to flat minima of L_η in $\eta = g^{-1}(\theta)$ and conversely. Figure 5 illustrates that point in one dimension. Several practical (Dinh et al., 2014; Rezende & Mohamed, 2015; Kingma et al., 2016; Dinh et al., 2016) and theoretical works (Hyvärinen & Pajunen, 1999) show how powerful bijections can be. We can also note that the formula for the transformed Hessian at a critical point also applies if g is not invertible, g would just need to be surjective over Θ in order to cover exactly the same family of prediction functions

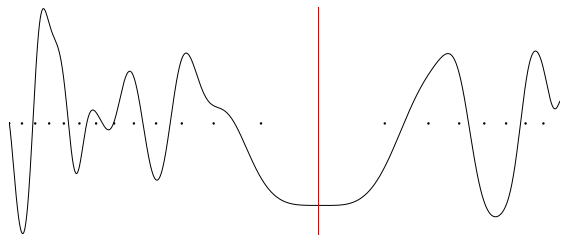
$$\{f_\theta, \theta \in \Theta\} = \{f_{g(\eta)}, \eta \in g^{-1}(\Theta)\}.$$

We show in Appendix A, bijections that allow us to perturb the relative flatness between a finite number of minima.

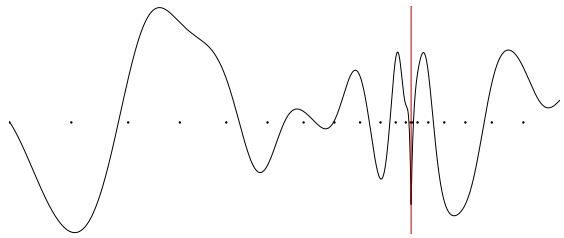
Instances of commonly used reparametrization are *batch normalization* (Ioffe & Szegedy, 2015), or the *virtual batch normalization* variant (Salimans et al., 2016), and *weight normalization* (Badrinarayanan et al., 2015; Salimans & Kingma, 2016; Arpit et al., 2016). Im et al. (2016) have plotted how the loss function landscape was affected by batch normalization. However, we will focus on weight normalization reparametrization as the analysis will be simpler,



(a) Loss function with default parametrization



(b) Loss function with reparametrization



(c) Loss function with another reparametrization

Figure 5: A one-dimensional example on how much the geometry of the loss function depends on the parameter space chosen. The x-axis is the parameter value and the y-axis is the loss. The points correspond to a regular grid in the default parametrization. In the default parametrization, all minima have roughly the same curvature but with a careful choice of reparametrization, it is possible to turn a minimum significantly flatter or sharper than the others. Reparametrizations in this figure are of the form $\eta = (|\theta - \hat{\theta}|^2 + b)^a(\theta - \hat{\theta})$ where $b \geq 0, a > -\frac{1}{2}$ and $\hat{\theta}$ is shown with the red vertical line.

but the intuition with batch normalization will be similar. Weight normalization reparametrizes a nonzero weight w as $w = s\frac{v}{\|v\|_2}$ with the new parameter being the scale s and the unnormalized weight $v \neq 0$.

Since we can observe that w is invariant to scaling of v, reasoning similar to Section 3 can be applied with the simpler transformations $T'_\alpha : v \mapsto \alpha v$ for $\alpha \neq 0$. Moreover, since this transformation is a simpler isotropic scaling, the conclusion that we can draw can be actually more powerful with respect to v:

- every minimum has infinite volume ϵ-sharpness;
- every minimum is observationally equivalent to an infinitely sharp minimum and to an infinitely flat minimum when considering nonzero eigenvalues of the Hessian;
- every minimum is observationally equivalent to a minimum with arbitrarily low full-space and random subspace ϵ-sharpness and a minimum with high full-space ϵ-sharpness.

This further weakens the link between the flatness of a minimum and the generalization property of the associated prediction function when a specific parameter space has not been specified and explained beforehand.

5.2 Input representation

As we conclude that the notion of flatness for a minimum in the loss function by itself is not sufficient to determine its generalization ability in the general case, we can choose to focus instead on properties of the prediction function instead. Motivated by some work in *adversarial examples* (Szegedy et al., 2014; Goodfellow et al., 2015) for deep neural networks, one could decide on its generalization property by analyzing the gradient of the prediction function on examples. Intuitively, if the gradient is small on typical points from the distribution or has a small Lipschitz constant, then a small change in the input should not incur a large change in the prediction.

But this infinitesimal reasoning is once again very dependent of the local geometry of the input space. For an invertible preprocessing ξ^{-1}, e.g. *feature standardization*, *whitening* or *gaussianization* (Chen & Gopinath, 2001), we will call $f_\xi = f \circ \xi$ the prediction function on the preprocessed input $u = \xi^{-1}(x)$. We can reproduce the derivation in Section 5 to obtain

$$\frac{\partial f_\xi}{\partial u^T}\big(\xi(u)\big) = \frac{\partial f}{\partial x^T}\big(\xi(u)\big)\frac{\partial \xi}{\partial u^T}(u).$$

As we can alter significantly the relative magnitude of the gradient at each point, analyzing the amplitude of the gradient of the prediction function might prove problematic if the choice of the input space have not been explained beforehand. This remark applies in applications involving images, sound or other signals with invariances (Larsen et al., 2015). For example, Theis et al. (2016) show for images how a small drift of one to four pixels can incur a large difference in terms of $\mathcal{L}_2$ norm.

6 Discussion

It has been observed empirically that minima found by standard deep learning algorithms that generalize well tend to be flatter than found minima that did not generalize

well (Chaudhari et al., 2017; Keskar et al., 2017). However, when following several definitions of flatness, we have shown that the conclusion that flat minima should generalize better than sharp ones cannot be applied as is without further context. Previously used definitions fail to account for the complex geometry of some commonly used deep architectures. In particular, the non-identifiability of the model induced by symmetries, allows one to alter the flatness of a minimum without affecting the function it represents. Additionally the whole geometry of the error surface with respect to the parameters can be changed arbitrarily under different parametrizations. In the spirit of (Swirszcz et al., 2016), our work indicates that more care is needed to define flatness to avoid degeneracies of the geometry of the model under study. Also such a concept can not be divorced from the particular parametrization of the model or input space.

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